ENGINEERING ECONOMICS

LECTURE - 11

Cost/Benefit Analysis

A systematic comparison of the expected costs and benefits of a course of action.

- When benefits and costs are measured on the same scale, such as dollars, the benefits should exceed the costs for a given course of action.

- When the alternatives are estimated to provide the same benefit, the alternative with the lowest cost should be selected.

Benefit Measurement Methods

Economic Models

The process of identifying the financial (economic) benefits is called **Capital Budgeting**.

It is the decision-making process by which some organizations evaluate and select projects.

Cost/Benefit Analysis

Benefit Measurement Methods

Economic Models

- Payback Period
- Discounted Cash Flow
  - Net Present Value
- Benefit/Cost Ratio
- Internal Rate of Return (IRR)

Payback Period

Payback period is the length of time, usually expressed in years or fractions there of, needed for a firm to recover its initial investment on a project.

For example, a $1000 investment which returned $500 per year would have a two year payback period.

An assumption in the use of payback period is that returns to the investment continue after the payback period.

Payback Period Example

Initial Project Expense = $5,000

Year	Amount Paid	Remaining Amount
Year 1	$1,000	($4,000)
Year 2	$2,000	($2,000)
Year 3	$2,000	$ 0

Payback Period (Criteria)

An organization that uses Payback Period would also have to define what the payback period criteria would be?

Some organizations would be very happy with a payback period of three years. Others would no doubt use a much shorter payback period criteria.

Future Worth and Present Worth Concepts

Future Worth

FW = PW (1+interest rate)

raised to the (number of years) power.

Example: Lets say we have $1,000 invested at 6% for three years.

FW = $1,000 (1+.06) to the third power.

FW = $1,000 * (1.1910)

FW = $1,191

Future Worth Table

Years	2%	3%	6%	10%
1	1.0200	1.0300	1.0600	1.1000
2	1.0404	1.0609	1.1236	1.2100
3	1.0612	1.0927	1.1910	1.3310
4	1.0824	1.1255	1.2624	1.4641
5	1.1040	1.1592	1.3382	1.6105

Present Worth

The result of discounting one or more amounts to be received or paid in the future by a *discount rate*.

PW = FW * 1 / ((1+interest rate) to the (number of years) power).

Example 01: $100 invested at 6% will amount to $106 at the end of one year (this is a future worth). Therefore:

The **present worth** of $106 due at the end of one year at 6% is $100.

Example 02: Lets say $1,000 being sent to us 3 years from now and the interest rate is at 3%. Calculate PW?

PW = $1,000 * 1/((1+.03) to the third power

PW = $1,000 * (.9151)

PW = $915.10

Present Worth Table

Years	2%	3%	6%	10%
1	.9803	.9708	.9433	.9090
2	.9611	.9425	.8899	.8264
3	.9422	.9151	.8396	.7513
4	.9238	.8884	.7921	.6830
5	.9057	.8626	.7472	.6209

Present Worth Analysis

It is the recognition that any amount due in the future is worth ***less*** than that same amount if it were due today.

Discounted Cash Flow

The present worth of all expected net cash receipts from a project, discounted by an appropriate discount rate.

Discount and Discount Rate

A "Discount" is a "Charge" that is paid to obtain the right to delay a payment.

The "Discount", or "Charge" that must be paid to delay the payment, is simply the difference between what the payment amount would be if it is paid in the present and what the payment amount would be paid if it will be paid in the future.

The discount rate

The rate used to discount future cash flows to their present values.

Discounted Cash Flow

Initial Project Expense = $5,000

(Payback) Discounted Cash Flow at 6%.

	Future Value	Present Value	
Year 1	$1,000	$ 943	($4,057)
Year 2	$2,000	$1,780	($2,277)
Year 3	$2,000	$1,697	($ 580)
Year 4	$2,000	$1,584	$1,004

Net Present Worth or Net Present Value

The algebraic sum of the **present worth** of all outflows and inflows associated with a given project or investment.

Calculation of net present worth usually involves subtracting the initial outflow cost of an investment from the present worth of all future cash flows.

Net Present Worth

Discounted Cash Flow at 6%.

Year 1	$1,000	$ 943
Year 2	$2,000	$1,780
Year 3	$2,000	$1,697
Year 4	$2,000	$1,584
Total		$6,004 accrued benefit
Less Investment		- 5,000
Net Present Worth		$1,004

Benefit/Cost Ratio

B/C ratio is actually a ratio of discounted benefits to discounted costs.

$$\textbf{B/C} = \frac{\$PW\ (\text{Benefits})}{\$\ PW\ (\text{Cost})}$$

$$\textbf{B/C} = \frac{\textbf{\$ PW (B)}}{\textbf{I+\$ PW (O\&M)}}$$

B=Benefits

I=Initial Investment

O&M= Operating and maintenance cost

$$\text{Modified B/C} = \frac{\textbf{PW (B)-PW(O\&M)}}{\textbf{I}}$$

Note: A project is acceptable when B/C ratio is greater or equal to 01.

Benefit/Cost Ratio

$$\frac{\text{Project Benefit} \quad \$\,7{,}000}{\text{Project Cost} \quad \$\,5{,}000}$$

Benefit/Cost Ratio = 1.4

An organization could establish any "criteria" that they wanted for the purposes of evaluating a project. Company A might have a Benefit/Cost Ratio requirement of 1.5 or greater. Company B might simply make the decision to do the project if it had a Benefit/Cost Ratio of 1.0.

Internal Rate of Return (IRR)

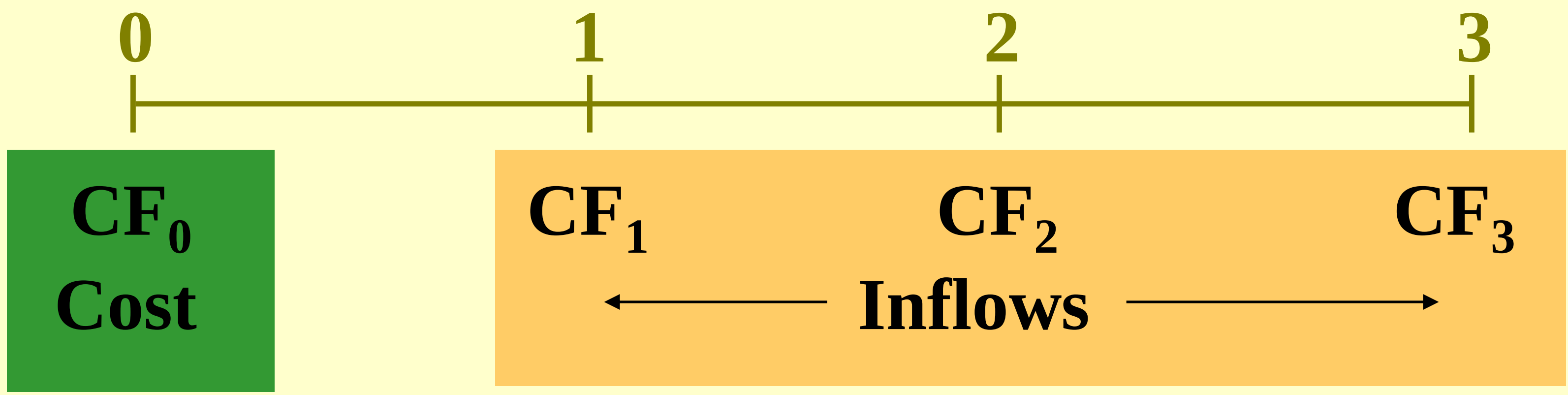
The discount rate often used in capital budgeting that makes the net present value of all cash flows of a particular project equal to zero.

Generally speaking, the higher a project's internal rate of return, the more desirable it is to undertake the project. As such, IRR can be used to rank several prospective projects a firm is considering.

Assuming all other factors are equal among the various projects, the project with the highest IRR would probably be considered the best and undertaken first.

IRR is sometimes referred to as "economic rate of return (ERR)".

Internal Rate of Return: IRR



IRR is the discount rate that forces PV inflows = cost.

This is the same as forcing NPV = 0.

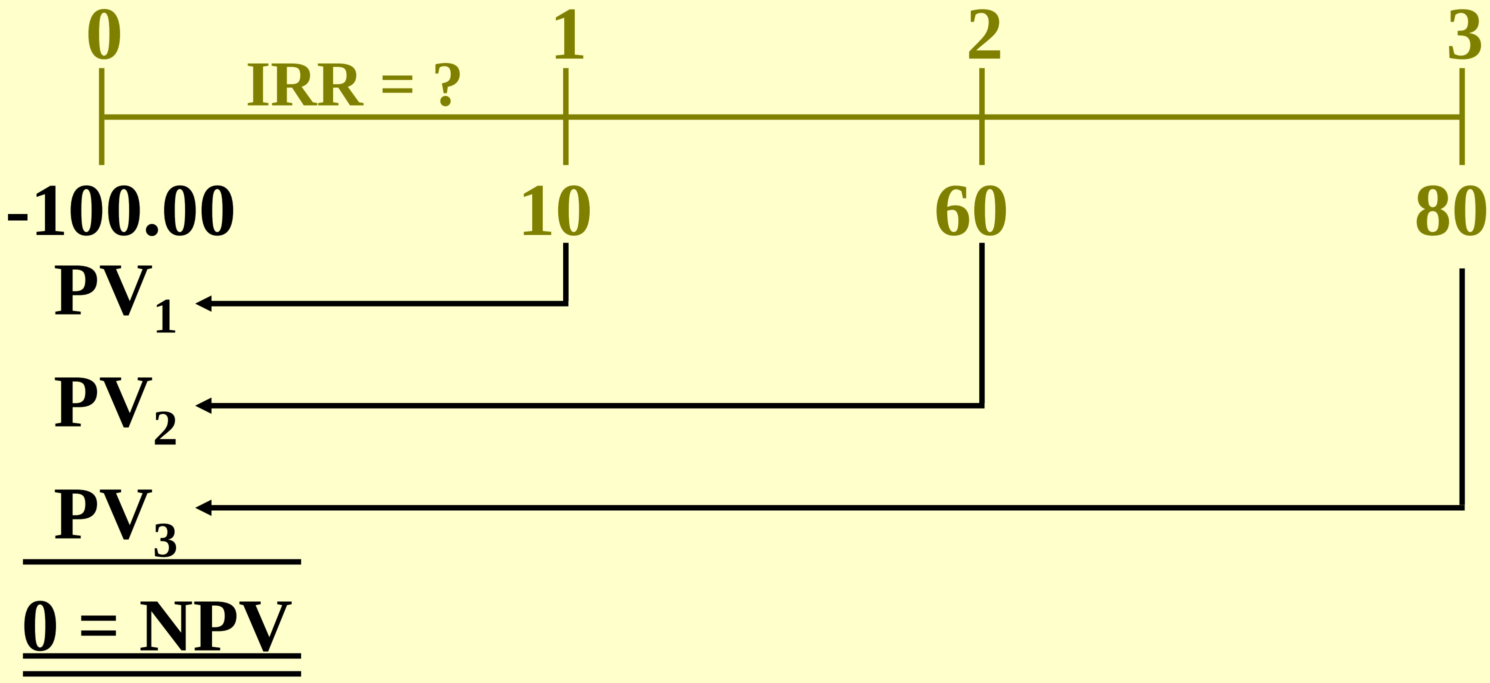
NPV: **Enter r, solve for NPV.**

$$\sum_{t=0}^{n} \frac{CF_t}{(1+r)^t} = NPV.$$

IRR: **Enter NPV = 0, solve for IRR.**

$$\sum_{t=0}^{n} \frac{CF_t}{(1+IRR)^t} = 0.$$

Calculate IRR



IRR = 18.13%.

Find IRR if CFs are constant:

0	1	2	3
-100	40	40	40

IRR = ?

IRR = 9.70%.

Discussion